

ADAPT Project ANR-23-CE25-0004

Data Aggregation on Modular Robots

Delivery Report D0.1.b

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December 2025

Abstract

The report presents the instrumentation-driven framework for measuring and monitoring energy consumption, delivering the Adapt report D0.1.b on data collection and data aggregation in modular robotic systems (MRs) represented by Blinky Blocks (BBs). Such cyber-physical systems are composed of identical modules that perform computations and communicate with their neighbors and the environment by using sensors and actuators. Based on MRs' metrics and parameters for energy-aware design, a custom designed low-power PCB¹ is first described that enables current sensing per module at high resolution. The PCBs are then integrated in a LabVIEW-based data acquisition pipeline capturing energy consumption across various module-based patterns. This instrumentation allows a high resolution tracking of actuation, sensing and communication phases when varying metrics and parameters. Afterwards, the collected data are stored in a database for processing with DataVIZ, an application to analyze and visualize them. A visual analysis tool mapping energy consumption profiles to execution timelines reveals critical information and provides useful insights for BBs, such as anomalous consumption patterns or inter-module variability.

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¹Printed Circuit Board

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1 Introduction

Energy management is a major challenge for cyber-physical systems [4], particularly in large-scale deployments where power distribution, propagation, and optimization must be carefully controlled. Hence the CPS development requires improved design methodologies and tools to support specification, modelling and analysis of both cyber and physical aspects [2]. Recent work [5] describes a methodology and a tool support to address this question. However, there are methodological difficulties and lack of data for building CPS' models, which must be trustworthy to allow a meaningful model-based analysis. Moreover, the CPS' 5C architecture in [1] contains a data-to-information conversion layer with the purpose to help the developers design better and more robust systems and equipment by leveraging data from operational performance. Indeed, converting these data to information may help data correlation for more precise specifications and meaningful models, on the one hand, and degradation and performance analysis and prediction, on the other hand.

In this report, we consider modular robots (MRs), a kind of CPS composed of modules that perform computations and communicate with their neighbors and the environment by using sensors and actuators. They interact to achieve a converging goal by cooperating. to convey small objects, to make some sounds and/or change their colour, to constitute some shapes while sliding, etc. We focus on Blinky Blocks [3] (or BBs for short) that offer a real playground for the programmable matter projects.² Identifying and understanding key metrics such as energy consumption (a physical aspect), or communication overhead and resource allocation (computer science aspects) is crucial for enhancing their efficiency, adaptability, and scalability. Also, for the identified metrics, data collection and correlation allow their subsequent integration into specifications and models, that become then trustworthy.

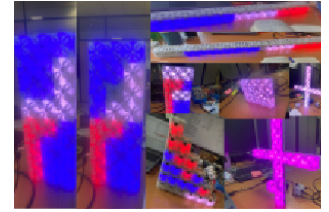


Figure 1: BBs motifs

Our work on BB's key metrics, data collection and models is strongly motivated by the practical need to avoid destroying Blinky Blocks while building complex BB-based systems composed of hundreds of components. Indeed, the traditional trial and error approach has already caused damage to several BBs. In addition, the MR developer can benefit from the collected data while programming BB-based robots, e.g., by avoiding LEDs set to the values leading to energy consumption peaks, blackout, etc.

2 Blinky Blocks Systems

2.1 BB-based Systems: Motivating Issues

Blinky Blocks systems are very useful for developing MRs algorithms and protocols. Indeed, while simulation [9] allows implementing distributed algorithms on millions of virtual MRs, MRs physical implementations rarely exceed dozens, or a few hundreds components. Blinky Blocks have been designed to be simple enough so that one may manufacture thousands of them. For our experiments, we have at disposal 2000 of them, which make a good intermediate solution to test real-life MRs at a scale in-between what is usually achieved with physical robots and what can be simulated. Thus, it enables us to test distributed algorithms with real hardware and software. It allows underlining hardware-related issues that cannot be simulated precisely, while still having a size sufficient to make large scale MRs systems observations.

Blinky Blocks also come with their flaws. They have been designed to be based on COTS³ components and a simplified architecture to be cost effective. As such, they have limited capabilities, especially when it comes to self-awareness. In the scope of energy management and optimization, this translates to robots unable to measure the quality of their power supply (current and voltage), which leads to setups with power supplies count and location based on (more or less) educated guesses. Already, some unintended risky configurations have lead to losses in our Blinky Blocks set: some completely burnt, others had some of their network interfaces melting. Thus, two main issues identified with the distributed power supplies are:

- **Malfunctions:** This issue is benign since it just prevents the set of robots of performing its task. Usual symptom is a subset of Blinky Blocks not performing correctly (no light or no sound). Adding power supplies and checking connectors alignment helps solving this kind of issue.

²as, e.g., the Pm project.

³Commercial off-the-shelf

- **Blinky Blocks destruction:** This issue, fortunately very uncommon, is critical since it may destroy hardware. It happens when too much current is served through a path, leading to a high Joule's effect that may melt components of the Blinky Block (up to the MCU). Usual symptoms are smoke and burning circuit smell. Solving the issue requires checking everything, add power supplies and replace the destroyed BBs. The larger and more complex the system is, the more likely the issue occurs.

2.2 Energy Metrics and Parameters

Modular robotics systems energy consumption does not solely depend on the drawn sum of per-block currents. Actually, it emerges from the interplay between multiple configuration like the topology, power supply units (PSU) distribution and command execution schedule. To capture this complexity in a formal yet generalized way, we introduce six core metrics, denoted (A-F). Each metric is experimentally grounded in our Blinky Blocks (BB) platform but can be extended to other modular robotic systems.

A) Motif (M)

The motif describes the structural arrangement of blocks and is modeled as a graph $G=(\mathcal{V},E)$, where the set of vertices $\mathcal{V}$ corresponds to the modules (BBs) and the edges E denote physical connections. Blinky blocks are aligned on a 3D cubic lattice. Motifs define the power distribution pathways as well as distances between PSUs and consuming modules. Typical motifs explored in our experiments include lines, rectangles, hollow rectangles, crosses, and three-dimensional extensions. The role of motifs in shaping scalability has also been emphasized in graph-based modular robotics literature [6, 10].

B) Number of PSUs ($|S|$)

We denote by $S \subset \mathcal{V}$ the subset of blocks connected directly to the external supply. The cardinality $|S|$ therefore represents the number of PSUs in a given configuration. Increasing $|S|$ distributes the load, potentially delaying failures and increasing the feasible system size. Conversely, a single PSU often becomes a bottleneck in motifs with large diameters. The importance of power supply distribution has long been noted in modular robot surveys [12, 11].

C) PSU Placement (S)

The placement of PSUs determines how voltage is distributed across the motif. Although the per-block current draw remains stable, blocks located farther from PSUs experience voltage drops due to series resistance in connectors and wiring. This reduces their effective power and may lead to under-performance or failure, particularly under concurrent high-load commands. We quantified this by the distance (in terms of hops) from any module i towards its nearest PSU:

$$X_i = \min_{s \in S} \text{dist}_G(i, s),$$

where dist_G is the shortest hops path distance in the motif graph. For example, in a line graph of 20 blocks with a single PSU at position 0, the block at position 10 has $X_{10} = 10$. A PSU placed centrally reduces the maximum distance compared to one at the edge, thus increasing tolerance to simultaneous loads. While most existing studies emphasize configuration and control architectures [7], our metrics highlight that PSU location introduces measurable differences in effective capacity and energy resilience.

D) Number of Blocks (N)

The total modules number in the system is $N = |\mathcal{V}|$. Based on N and the actions of the Blinky Blocks, we observe two types of configurations: the first one is linear and corresponds to power requirements less than the PSUs capabilities. The second one is non-linear and is observed when approaching or exceeding PSUs' capabilities. Such size-dependent constraints and scaling effects have been reported in modular robotics energy profiling[8].

E) Commands of the components (p)

Each BB module is required to execute a command $p \in P$ which translates into a power (e.g., LED colors and brightness, speaker buzzing frequencies). For LEDs brightness and colors, and Speakers frequencies, we rely on empirically measured data per-block. Based on our observations, we also denote that having the highest power demanding commands closest to the PSUs increases the ability of the system to operate as intended.

F) Command Order (π)

This temporal order (π) specifies the commands timeline for each Blinky Block. Commands can be simultaneous (several different actions on one BB, e.g. buzzing and lighting at the same time), or sequential (e.g. buzzing, then lighting). Simultaneity also exists at the global level: each BB executes at least one command at the same time. Simultaneous commands make BBs concurrent for accessing power.

2.3 Towards Fine-grained Measurement

Based on our preliminary observations, performed with multi-meters, we noticed lots of phenomena hard to explain with global and static measurements. In particular, multi-meters cannot account for transitions during the tests (powering up devices, changing the commands, etc.). For instance, when starting a whole configuration at once, or building it gradually, results differ (the first case will fail, while the second one will operate as expected). We have hypotheses such behavior, that need to be further explored. That is why we developed a more precise measuring platform.

Our measurements platform is based on the following elements:

- PCBs with a kind of socket to plug a BB into. Each PCB embeds a sensor to measure the current drawn by its BB,
- An acquisition system, based on LabVIEW, to sample the sensors readings at high rates,
- A database to collect the data,
- An application to analyze and visualize the data, in order to derive power models from the data collected.

This platform enables us to see some transitions, since it can sample values in the order of the kHz.

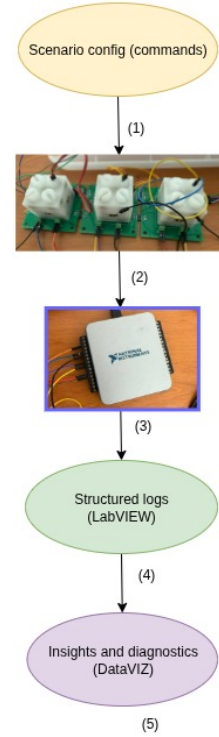


Figure 2: Overview

3 Framework for Data Collection and Management

The measurements platform we designed is part of a process, illustrated by Fig. 2. To support this process, we designed tools, composed of both physical as well as software elements, as follows:

- Observed subjects: Blinky Blocks executing a scenario (i.e. a set of commands) in a given motif,
- Measurements tools: PCBs, multi-meters, or a mix of both are used to get local and global metrics,
- Data storage: from a scenario, data measured is structured into a relational database,
- Visualization: a web-based tool is plugged on the database in order to draw various graphs of the data.

This process is embedded into a larger framework to provide self adaptation. The whole framework is composed of the following parts, that can be instantiated in various ways:

- Data acquisition: empirical power (voltage along with current) measurements,
- Power consumption modelling: for individual as well as global BBs set,
- Adaption strategy: offline or online segments of code added into the BBs' programs. The adaptation by itself may use various sub-modules, for instance to provide real-time feedback to the BBs.

Our current platform provides instances for the two first parts of the framework. In this section, first, we detail the PCB we designed for BBs, second, we present our data acquisition chain. Third, we describe what a scenario can be.

3.1 PCB Description

To improve data accuracy, we designed a custom PCB, shown in Fig. 3). It records real-time power variations for a Blinky Block plugged into its socket. The PCB includes a INA180 current sensors providing measurements to a DAQ peripheral connected to a computer running LabView. In order to keep measurements accurate, the PCB's own instrumentation (i.e. the current sensor) is powered from an external source so it does not add to the BB's consumption.

The board is designed with respect to a Blinky Block vertical chain: The bottom side of the BB receives power from its neighbor—through the PCB—while the top side provides power to the next Blinky Block's bottom side in the chain. We intend to design newer versions of the PCB to enable 2D motifs with 2 inputs and 2 outputs, instead of the current ones with 1 input and 1 output.

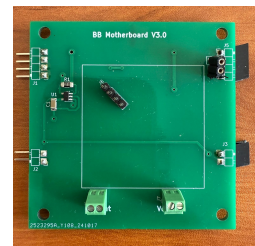
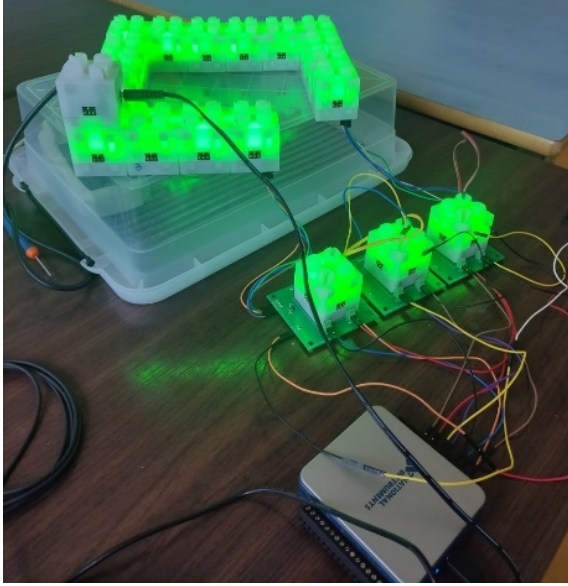


Figure 3: Designed PCB

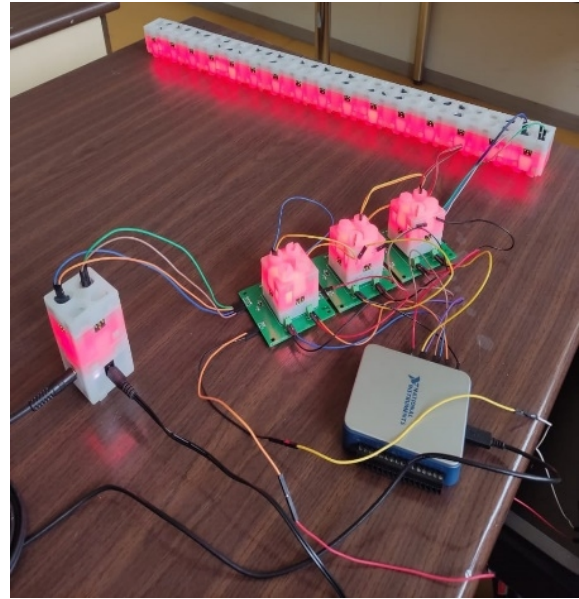
3.2 Data Acquisition Chain

This part of our experimental set makes the analog digital conversion of the INA sensor output, timestamps the sample, and writes it into a human readable file, similar to the CSV format. The data acquisition system (DAQ) is based on a National Instruments DAQ module, which allows the acquisition of synchronized current and voltage signals from our custom PCBs. Channels are used in parallel to capture the individual Blinky Block measurements samples.

Figure 4 illustrates the use of PCBs connected to the NI DAQ, with 2 different motifs: a hollowed square one, in subfigure 4a, in high green (RGB code 0,100,0), and a line motif in subfigure 4b, in high red (RGB code 100,0,0). Although the physical shape is partly dismantled to fit the PCBs in the motif, the electrical, network and logical shape remain unchanged.



(a) Instrumentation in a hollowed square pattern



(b) Instrumentation in a line pattern

Figure 4: Setup of the full instrumentation with different patterns and loads

Figure 5 illustrates a LabVIEW model used to implement the data acquisition and logging workflow. The DAQ Assistant module initializes the NI data acquisition card and streams current/voltage signals from the PCB sensors. These signals are then passed through a signal disassembly block, which separates individual channels corresponding to different Blinky Blocks. Each channel can simultaneously be: (1) displayed in real time using the Graph block for monitoring, and, (2), logged to file using the Write to Measurement File (.lvm) block for structured storage.

Control inputs such as sampling frequency and enable/disable toggles allow flexible configuration of experiments. Thereby, all measurements are ensured to be acquired synchronously across multiple channels, visualized in real time, and stored in a format compatible with our DataVIZ platform for post-processing and further analysis.

3.3 Scenarios

Scenarios are sets of timed commands for Blinky Blocks. Since each command will require some amount of power, this enables us to make scenarios for testing our hypotheses and theoretical models against real life measurements. Measurements are done on a set of BBs, arranged in a given motif. During an experiment, the tested BBs all execute a program, that may be identical for all, or have different commands (for instance, part of the BBs light in red, while others light in yellow). A scenario also defines how many PSUs are used, as well as their location in the motif.

At the software level, a scenario is a sequential set of commands for each component of a Blinky Block. It is sequential for a component since LED can only make one color at a time, or buzzers can only make one sound at a time. Parallelism of the commands still is possible by activating various components in the same time. The sequence of commands translates into C language in the BB. An example of a simple scenario is illustrated by listing 1.

```
1 void BBinit() {  
2     set_RGB(5, 0, 0);    // Low brightness red LED  
3 }  
4
```

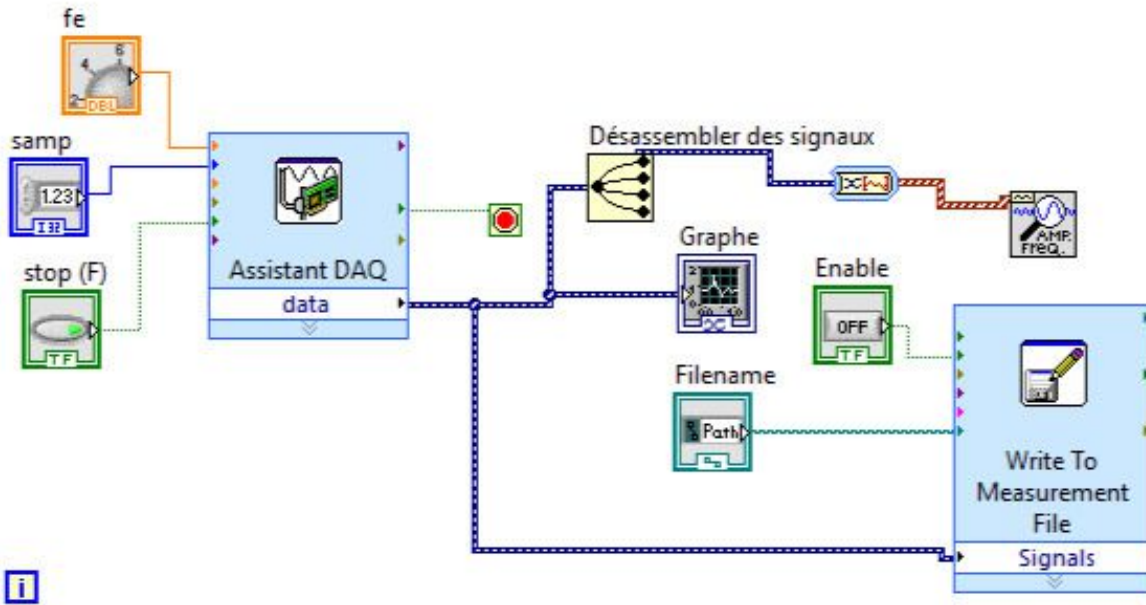



Figure 5: A LabVIEW model

```

5 void BBloop() {
6     sleep_sec(10);           // Wait for 10s
7     make_sound(440, 7000);   // Play 440 Hz tone during 7s
8     sleep_sec(7);           // Wait 7s
9 }

```

Listing 1: Scenario code for Blinky Block System

In this example, the Blinky Block will start with a dark red color, that will remain during all the test, and it will alternate 10 seconds silences with 7 seconds LA tone.

4 Experiments

We carried out several experiments for getting real-life data as a base for our consumption models. In the previous phase of ADAPT, we started with comparative individual static measurements on one Blinky Block to refine our theoretical model. Such measurements were performed with multi-meters that reported stable values. These values were manually stored into files describing the experiments and their results.

With a PCB designed to get high frequency measurements, we made measurements on Blinky Blocks within motifs, with changing commands. In the remainder of this section, first we describe our experimental procedure. Second, we provide results for an individual Blinky Block. Third, we show the results on motifs.

4.1 Experimental Protocol

Each experiment is carried out with the following procedure:

1. Parameters definition: the number of BB, their motif, their commands, the number of PSUs and their location are decided.
2. The commands are implemented. Currently, it consists in writing a program for BBs, then deploying it on the BBs
3. BBs are plugged on the measurements platform: it includes connecting the BBs to form the defined motif, including the PCBs located on measurements points.
4. The system is started: LabView capture starts, then the BBs' program.
5. During the run, values are logged by LabView, and multi-meters values are stored manually in a result file.
6. At the end of the experiment, everything is stopped, results are gathered are put in the DB, associated to a meaningful name (making motif, numbers, etc. explicit) and the date and time of the experiment.

After the experiment is performed and its results stored, measurements can be used in the offline phase addressing the modelling, which is out of the scope of this document. Since a set of BBs is a complex system, its behavior from the outside is not deterministic. To address this issue, we repeat each experiment several times.

Experiments on a single BB can be viewed as a specific motif with only one Blinky Block, like a "point" motif.

Our experiments have several objectives:

- Compare results with a PCB to results obtained with multi-meters,
- Estimate voltage drop while varying the number of Blinky Blocks in the system
- Estimate failure probabilities according to the number of Blinky Blocks and their commands,

4.2 Individual Blinky Block Measurements

This experiment aims at collecting data for a single Blinky Block with our custom PCB as well as providing an empirical consumption model from our observations. This work was carried out in the context of a master 2 internship. It produced a second degree polynomial regression model expressing consumption according to LED color's red, green and blue components.

The model uses normalized variables for modelling its color components:

$$R' = \frac{R}{100}, \quad G' = \frac{G}{100}, \quad B' = \frac{B}{100}$$

where R , G , and B represent the commanded color intensities on a 0 to 100 scale.

We define the estimated current consumption eC as:

$$eC = \beta_0 + \beta_1 R' + \beta_2 G' + \beta_3 B' + \beta_4 R'^2 + \beta_5 G'^2 + \beta_6 B'^2 + \beta_7 R'G' + \beta_8 R'B' + \beta_9 G'B'$$

This model was trained using the complete set of current measurements obtained for all RGB combinations, applying ridge regularization to prevent overfitting:

$$\beta = (X^T X + \lambda I)^{-1} X^T y$$

where:

- X is the feature matrix of normalized RGB values,
- y is the vector of measured currents,
- λ is the regularization coefficient,
- β contains the fitted parameters.

The resulting coefficients are:

$$\begin{aligned} \beta_0 &= 24.58, & \beta_1 &= 23.65, & \beta_2 &= 10.31, & \beta_3 &= 20.75, & \beta_4 &= 14.05, \\ \beta_5 &= 55.12, & \beta_6 &= 24.26, & \beta_7 &= 23.42, & \beta_8 &= 18.91, & \beta_9 &= 14.58, \end{aligned}$$

Based on this model, we can predict a Blinky Block's consumption for a given LED action defined by its color components. This is a key step into providing actionable parameters to a set of Blinky Blocks aiming at controlling their actions for achieving an adaptive system. In the robotics **sense-think-act** loop, this model provides data upon which to base the **think** step.

In particular, each blinky Block can use the model to predict its future consumption, enabling it to possibly lower its parameters values (the color components) to fit within a global power budget.

We also took the opportunity to compare the measurements obtained with the PCB to those obtained with the multi-meters, as showed in table 1, an excerpt of our data. This excerpt shows a peculiar value: the yellow (10, 10, 0) displayed by the bootloader when starting the Blinky Block. We see similar results with both measuring tools except on full (100%) blue.

Table 1: LED colors: measured current per block (mA). Ranges are min–max over repeated trials.

Color (R,G,B)	Multimeter (mA)		PCB (mA)	
	min	max	min	max
Red (100,0,0)	61.6	62.6	63	65
Blue (0,0,100)	83.9	86.0	63	65
Blue (0,0,50)	41.0	41.0	38	43
Yellow (10,10,0)	30.0	31.0	32	32

4.3 Voltage Drop Estimation Model

To approximate voltage distribution along a chain of Blinky Blocks (BBs), an empirical model was derived from experimental measurements. The voltage at the block directly connected to the power supply (V_{BBPs}) depends on the total number of blocks in the system (NBBs) as follows:

$$V_{BBPs} = 5.1773 - 0.01079 \cdot NBBs \quad (1)$$

where:

- V_{BBPs} — voltage at the power-supplied BB,
- NBBs — total number of BBs connected in the chain.

The voltage deviation per block (d) between the power-supplied BB and the farthest BB is then computed as:

$$d = \frac{(V_{BBPs} - V_{FBB})}{(\text{PosPS} - \text{PosFaBB})} \quad (2)$$

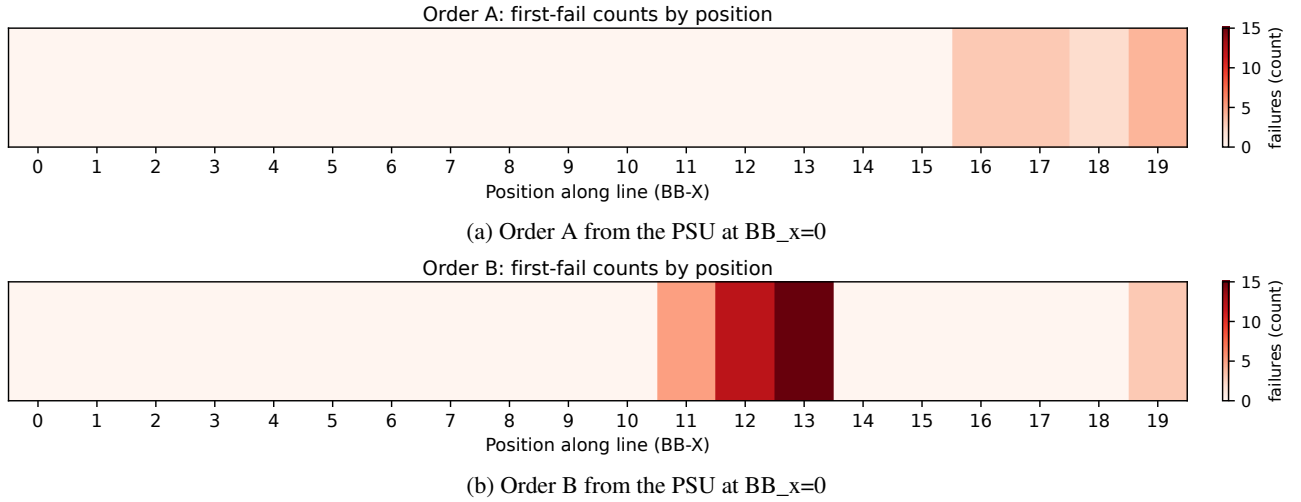


Figure 6: First-failure positions in the line motif (20 BBs) for two RGB command orders: Failures cluster farther from the PSU under order B.

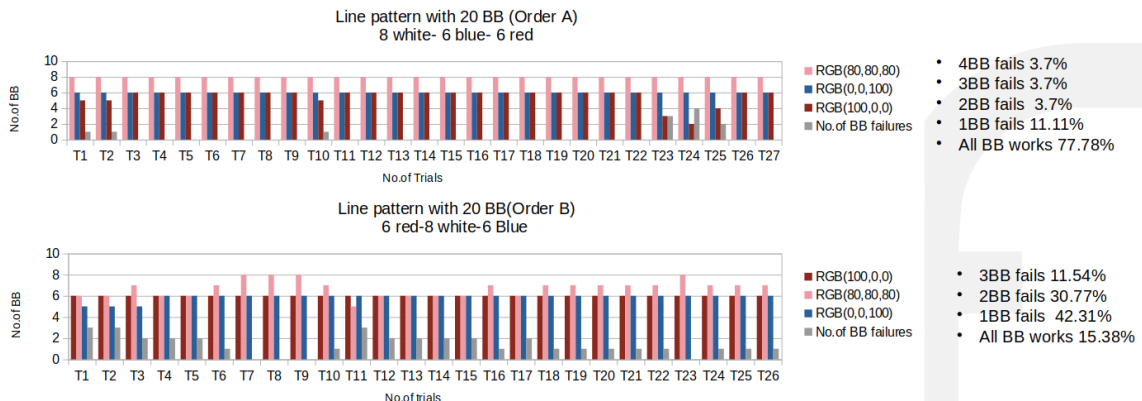


Figure 7: Line pattern of 20 BBs: Order of RGB commands w.r.t. the PSU

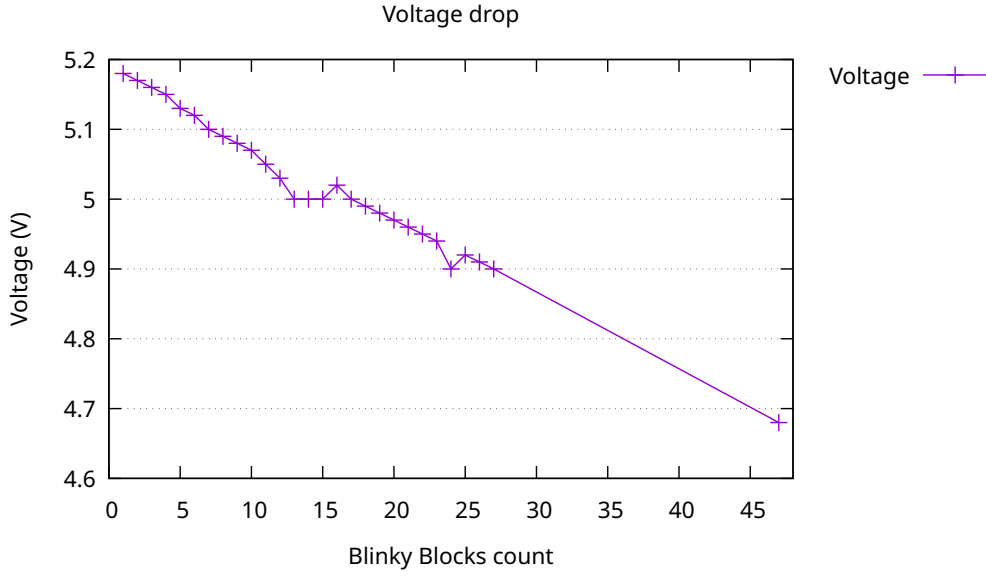


Figure 8: voltage drop of the central block according to the number of Blinky blocks connected (line configuration)

where:

- V_{FBB} — voltage at the farthest BB from the power source,
- $PosPS$ — position index of the power-supplied BB,
- $PosFaBB$ — position index of the farthest BB.

Using this deviation, the voltage estimated at any BB position (V_{EABB}) is given by:

$$V_{EABB} = (V_{BBPs} - (mD \cdot |PosABB - PosPS|)) + 0.1 \quad (3)$$

where:

- mD — average voltage deviation between blocks,
- $PosABB$ — position of the BB being analyzed,
- 0.1 — small correction factor derived from calibration.

Finally, the estimated current consumption of a single BB (CE) can be determined using Ohm's law:

$$CE = \frac{V_{EABB}}{R} \quad (4)$$

where R represents the equivalent resistance of one BB.

All calculations were implemented in a MATLAB graphical interface for rapid estimation of per-block voltage and current, as shown in Figure 9. This interface is based on multiple methods followed as Inverse Distance Weighting⁴ and Tikhonov regularization⁵. This interface supports design-time analysis of voltage decay and power safety thresholds across modular robot configurations.

4.4 Measurements for Blinky Block Sets

Figure 7 shows a comparison between two tests, which are repeated several times to gather statistical data, and build a model sound enough for offline energy optimization. We perform 26 or 27 trials for each test. From these samples, we compute the probability of block failure.

In this particular test, we keep the number of Blinky Blocks static, as well as the PSU placement. We vary the commands executed by the robots: the color partitions are swapped from 6 red-8 white-6 blue out of 20 BBs (bottom

⁴Method mentioned in Appendix section 8.1.1

⁵Method mentioned in Appendix section 8.1.2

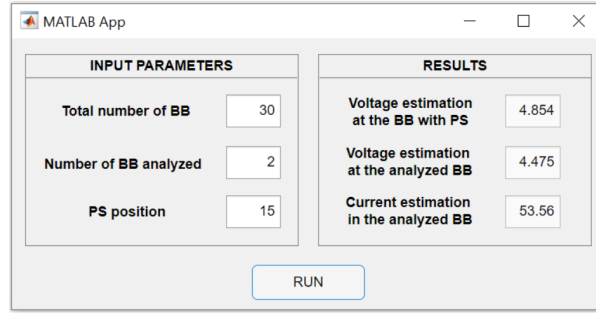


Figure 9: MATLAB interface for estimating voltage at a BB

plot), to 8 white-6 blue-6 red (top plot). The figure shows, for each test instance labeled as TN with N ranging from 1 to 26 or 27, the number of Blinky Blocks lit, with its color, and the number of failures. Failures are Blinky Blocks that cannot light their LED due to lack of power.

Figures 6 and 10 show other metrics correlation with the same configuration. Figure 6 indicates the blocks that fail the most w.r.t. the command order towards the PSU connected at $BB=0$. The figure points out the significant difference between the different orders, even if for both A & B orders, the last block always has a higher percentage to fail.

From both tests, we can deduce that higher energy demanding commands fare better when closer to the PSU, since we see only failures at the end of the line with the white-blue-red configuration, while the red-white-blue configuration gets failures close to its middle, where Blinky Blocks are white. Indeed, white is the most demanding LED command (about three times as much as red), and this combines with the cabling resistance to lose power faster, thus having more chance to lack power. We also observe that the end of the line is a little less faulty with the red-white-blue configuration, which is easily explainable, since prior failure on the line "spares" some power for the its remaining part.

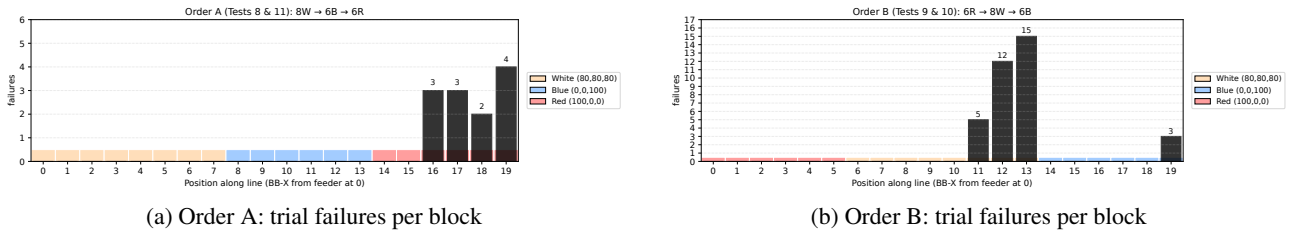


Figure 10: Failure numbers per block in the line pattern experiment.

Figure 10 depicts a different view on the failures for the same configurations. The lower part with each block bands and the black bars show BB's failure number out of the trials total number per order (A & B). It provides an indication on the command assigned to a group of blocks that had shown the highest failures and anomalies.

4.5 Observations for BB-based Motifs

This section describes the observations of BB-based motifs composed of up to 20 Blinky Blocks with a full white color, and a PSU on the edge of the motif. Motifs studied are: the line, the hollowed square, the filled square, and the star (which is better defined as a cross). The used PSU is a 5V, 2A PSU.

Figure 11 illustrates the measured current for these motifs, while varying the number of Blinky Blocks. We notice several interesting phenomena:

- The measured current can exceed 2A. Since we did not measure voltage at the same time, we think it probable though not certain, that the PSU compensates with a lower output voltage, thus not exceeding too much its design 10W power.
- We also denote decreases in measured current, which are actually matching failures to light on, meaning that not every Blinky Block is consuming its intended power.
- More generally speaking, the graph verifies the almost linear increase of current used by the Blinky Blocks according to their number. This was expected since each Blinky Block consumes the same for a given command, thus adding Blinky Blocks adds a fixed amount of power consumption.

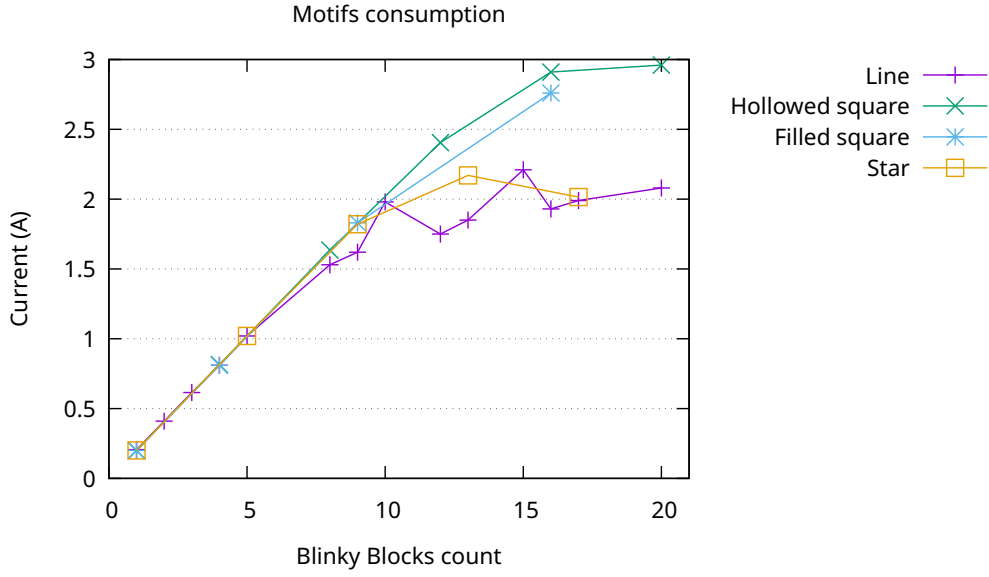


Figure 11: Patterns consumption for RGB(100,100,100) with 1 PSU at the edge

5 Instrumentation Integration

We described our experimental setup as well as some of our results. This work has been performed in the scope of a full framework going from experiments design - motif, size, etc. - to adaptation policy modeling. This requires several intermediate steps:

- Running the experiments, and obtain measurements: this step is carried out through physical implementation of the motif with Blinky Blocks , and embedding our custom PCBs into the motif. Measurements are retrieved by LabView,
- Gathering the data,
- Analyzing the data,
- Model power consumption of Blinky Blocks motifs,
- Design adaptation policies to prevent failures in running scenarios.

While we dealt with the first step in this report, we also have started implementing the remaining steps, as well as the measurements platform's integration into this framework. In this section, we briefly describe how it is implemented.

5.1 Data Collection & Analysis

After getting formatted data from LabView, we chose to build a web-based application, which can store the data, as well as visualize it. The application stores experiments parameters such as motifs, with individual Blinky Blocks identifiable. Blinky Blocks in the experiment's record are associated with their measurements file.

Data can be accessed through a web UI which allows modification, creation, deletion and visualization of the experiments' data. In particular, the application allows to graph time series values read and saved by the NI DAQ.

Figure 12a shows a Blinky Block motif in line, with 20 Blinky Blocks . Clicking on one module in the line allows opening its properties, among whose the time series measurements. The later can be displayed in a graph like the one depicted in fig 12b. By conducting experiments, then visualizing the results into this Web UI, we can get hints of the adequate modelling for power consumption.

5.2 Adaptation Policy

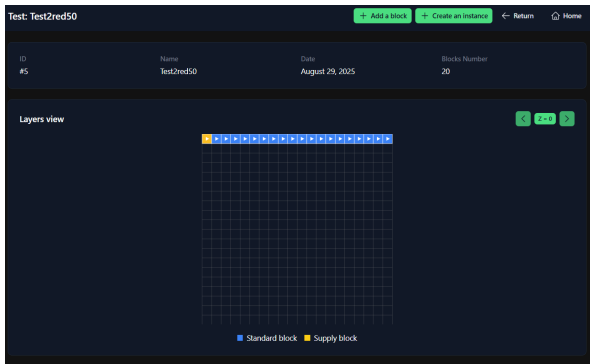
Our preliminary work on this step has only tackled offline adaptation: We design a scenario for an application. Based on our power consumption modelling, we factor the power budget in. If our scenario exceeds or is too close to the power budget, we reduce the commands' power requirements. As for now, this can be done with two strategies:

- For LEDs: we reduce the intensity of the required light, e.g., a requirement for a full white (RGB=100, 100, 100) would be tried with a lesser white (RGB=80, 80, 80).
- For sound: we select a subset of the Blinky Blocks that must make the sound, e.g. 2 out of 3, so that we reduce the power requirements from a value equal to that of the Blinky Blocks removed from the command.

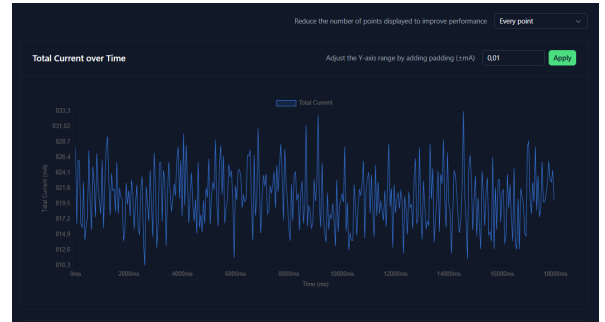
For online adaptation, work is still ongoing. The major issue is the lack of self-analysis in Blinky Blocks, meaning they cannot sense their own electrical supply. Since they can compute their modeled power consumption based on their commands, our task is now to design a method for providing feedback to the Blinky Blocks.

We already thought about two potential solutions: in the short term, we can hardcode the power budget inside our applications, and let Blinky Blocks act and adapt with their power consumption model. This is only a short term degraded solution, since it relies only on a model, without any feedback from a context where the model might be wrong. The second solution, to a longer term, requires instrumenting the programming blocks (the ones used to connect the power supply) with a chip that can interact with the Blinky Blocks. Before making dedicated designs, a first step towards this goal can simply use Arduino-like boards, connected to the programming block, and sensing voltage. This is a first quick and dirty solution, with its merits: since a high load on the PSU translates to a voltage drop, and since the Arduino boards can directly sense in the 0-5V on their pins, it can be achieved with a couple wires, and an Arduino application that implements the Blinky Blocks network stack.

This type of instrumentation would enable us to start programming adaptive applications, and to tackle more dynamic situations, ranging from a simple command program, to a varying number of Blinky Blocks with complex commands combinations.



(a) A visualized line pattern of 20 BBs



(b) DataVIZ: Pattern total consumption plots

Figure 12: DataVIZ platform

6 Conclusion

This deliverable report describes the contributions obtained in order to evaluate and optimize key metrics for modular robots, specifically Blinky Blocks (BBs). These robots, designed for collaborative tasks, require careful assessment of performance, capability, and system efficiency to ensure they meet the specific needs of various applications. The ADAPT project's objective to model and control adaptive distributed systems relies on measuring and optimizing physical and computational aspects of these systems.

The document details the energy consumption data collection and aggregation for BB-based systems, highlighting the necessity of accurate energy management to prevent anomalies. By providing an instrumentation pipeline of design, experiments and visualization. This work is crucial for advancing the understanding and efficiency of modular robot systems. Indeed, we succeed to analyze Modular robots behavior w.r.t. energy metrics when using this pipeline and varying the key parameters identified, e.g., the number of robots, their load, the power supply location in the BB pattern, and the layout of loads (if unequal) within the pattern.

Overall, the conclusions drawn from this study underline the critical role of energy consumption estimation and control in the development and optimization of modular robot systems, contributing to the broader goals of the ADAPT project. Moreover, this delivery report describes the work performed to address open questions from the previous report D0.1a future work intentions, namely on a tool chain for controlling BBs ensemble with their energy requirements shown in our pipeline instrumentation here. We also added more data, especially in terms of studied patterns: instead of investigated 1D patterns, we tried 2D patterns (e.g., rectangles) obtaining models suited to a more realistic usage. For

the moment, we are undergoing instrumenting the power supplies, and every connection in a pattern. According to the granularity and the extent we can reach that condition the volume of data we can acquire.

Further steps will consist in working on three main tasks. First, we aim to acquire a full model representing the whole interplay between all metrics of energy behavior. Second, we plan to study and to model the communication and messages exchanged between the BBs in the system. Last, studying MRs needs not only understanding and gathering data, but also modelling all the features MRs systems consist of, and applying adaptation and control policies upon them. This is one of the target objectives of the ADAPT project.

7 Acknowledgment

We would like to thank Silvio Vescovo and Félix Lahaut for their valuable work on the DATAVIZ project, which enabled a structured and insightful visualization of the gathered data. Their contribution was instrumental in presenting the results in an oriented and accessible way. We also wish to thank Alejandro Solis Rodríguez for his significant efforts in data aggregation and his support in interpreting the system's behavior under different configurations. His analytical input greatly contributed to understanding the overall performance and dynamics of the system. Finally, we acknowledge the appendix prepared by Anna Gallone with the explanations on the Matlab methods used by Alejandro Solis Rodríguez for energy consumption approximation.

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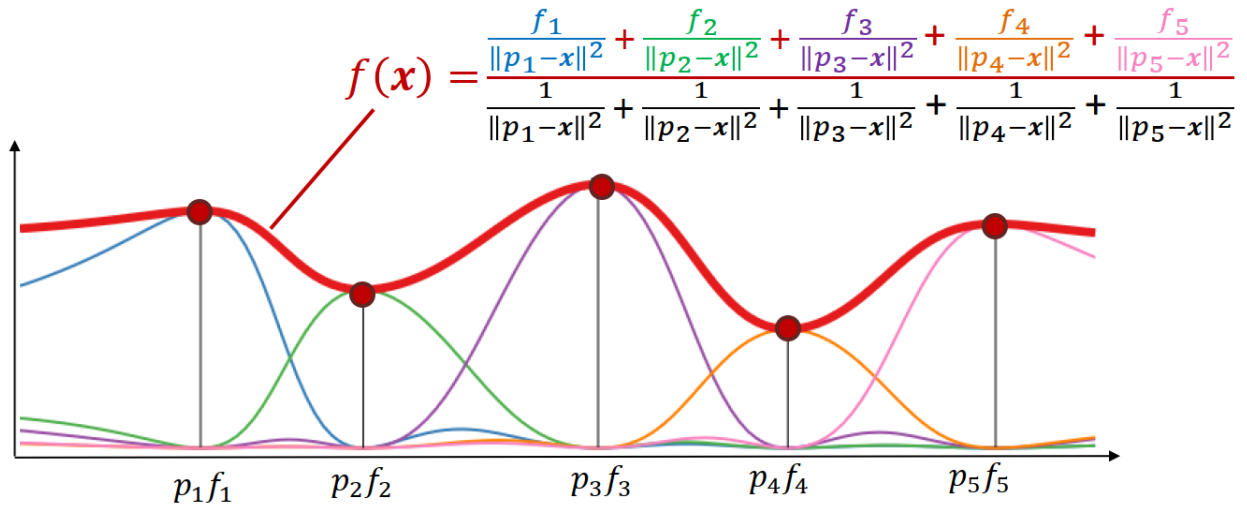


Figure 13: Graphical representation of the influence of weights attributed to samples of an Inverse Distance Weighting interpolation of a 1-D function

8 Appendix

8.1 Matlab Interface Methods

8.1.1 Inverse Distance Weighting

To provide an approximation of the current for any color, the first method is the Inverse Distance Weighting (IDW) Method. This method consists in building a model that takes into account every sample of couple in $(x_i, y_i)_{i \in [1, n]}$ (n being the number of samples) at disposal to predict the value of y according to a new x (see Fig. 13, by I.Elgamal - CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=137323597>).

In our case, x s are the values of R, G and B on a scale from 0 to 100 each. y s are the values of the current consumed by the blocks for such RGB configuration. Of course, according to the input x given, not all (x_i, y_i) will have the same influence. The closer the new input x is to an x_i , the bigger the influence of y_i over y . Since the report is a little inconsistent in the use of variables. A corrected (hopefully clearer) version of the equation for the current estimation $\hat{I}$ according to the collected measurements $(rgb_i, I_i)_{i \in [1, n]}$ is proposed :

$$\hat{I}(x) = \begin{cases} \frac{\sum_{i=1}^n w_i(x) \times I_i}{\sum_{i=1}^n w_i(x)} & \text{if } \forall i \in [1, n], x \neq rgb_i \\ I_i \text{ with } i \text{ such that } rgb_i = x & \text{otherwise} \end{cases}$$

having

$$w_i(x) = \frac{1}{\|x - x_i\|^p}$$

where

- $\hat{I}$ is the current estimation of the model (in Amperes),
- x is a triplet of values for R, G and B (in $[0, 100] \times [0, 100] \times [0, 100]$)
- w_i is the weight accorded to a sample (rgb, I) in the estimation (no unit),
- $\|\cdot\|$ is the Euclidean distance function,
- $n \in \mathbb{N}$ is the number of measurements taken into account to make the estimation and
- p is an arbitrary parameter with no unit to tune the weight given to different samples influence on the estimation.

In the report, since there is no case distinction made when the sample is already known, the weight given in such situation is set to 10^{12} . This arbitrary value can seem big enough to render other weights negligible in comparison, but in some situation, this discontinuity makes the estimation unable to retrieve the current correctly. Namely, if two rgb

measures x_1 and x_2 have a distance smaller than 10^{-4} separating them (in the case of $p = 3$ as in the report), than the I_2 value has more weight than I_1 when estimating the current for x_1 .

Anyway, in our situation, it is fine because our data for the current only used integers for the colors tested. so the minimal distance between two values of R, G and B would be 1.

8.1.2 Tikhonov Regularization for Polynomial Regression

The second method is polynomial regression. To choose the coefficients of this estimation technique, the Tikhonov regularization is used. To find an estimation of an output according to the data gathered earlier, this technique assumes that there is a linear relationship between all the possible input values (combinations of r , g and b and the output value (the current consumption of the block). Under this assumption, there must exist a set of $n + 1$ coefficients β_i that multiply the n inputs taken into account (+1 for adding a constant, $input_0 = 1$) to provide an estimated current eC :

$$eC = \sum_{i=0}^n input_i \times \beta_i$$

To find such coefficients, we must care about all the samples at the same time and find the set of β_i that suits best all of them. So we try to solve the β_i for :

$$Y = X\beta_{mat} + \epsilon$$

where :

- Y is a $k \times 1$ matrix containing all the electrical currents measured for the k samples,
- X is the $(n + 1) \times k$ matrix containing the k sample inputs that were used to measure the k electrical current values,
- β_{mat} is the $(n + 1) \times 1$ matrix containing the β coefficients
- and ϵ is the error that should be as close to zero as possible.

To find such set of β_i , the Tikhonov regularization uses operations on matrices the same way one would solve the problem with Ordinary Least Square (OLS) method. However, to avoid a too important sensitivity to small changes in the input values, instead of using directly the *moment matrix* ($X^T X$) in the expression of β_{mat} like in the OLS method :

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

This regression introduces a bias of $\lambda \in \mathbb{R}^+$ on the diagonal of the moment matrix which gives the following estimator:

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

with I being the identity matrix.